Comparative Analysis of Machine Learning Models for Stock Price Prediction

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***Abstract*— This paper conducts a comparison of random forest, GBM, and LSTM models in stock price prediction based on the data from Yahoo Finance. Random Forest and GBM have been the best among the models compared in this paper, and Lag_5 and MA_50 are found as the most important features. In contrast, LSTM performed poorly because it lacked enough data. This may mean that in some cases, easier models can give more reliable forecasts.**

I. Introduction

Correctly predicting stock prices is very important in making investment decisions and planning finances. This study comparatively analyses three popular machine learning models—Random Forest, Gradient Boosting, and Long Short-Term Memory (LSTM)—in forecasting stock prices based on historical data from Yahoo Finance. Comparing the performance of each model from the indicators such as Mean Absolute Error, Root Mean Square Error, and R-squared value, the research is aimed at finding the most dependable and efficient method out of them for the prediction of market trends. Findings can provide insight into the selection of financial models for analysts and investors looking to apply machine learning to improve the support of better decision making.

1. Literature Review

A. *Theoretical Background*

Stock market behavior remains difficult to accurately predict due to the dynamism and volatility of financial markets, and the endless number of factors that cause these sometimes-random fluctuations. Generally, traditional statistical models, such as the stock price data, are driven by nonlinear relationships and complex patterns (Box et al., 2015). Machine learning provides a robust alternative by applying algorithms that can model complex interactions within large datasets. Some of the important ML approaches are, in fact, supervised learning techniques that include such algorithms as regression models, decision trees, and neural networks, which attempt to predict future values based on historical data. It also enhances predictive accuracy, as reinforcement learning optimizes decision-making processes in trading environments. These theoretical foundations justify the increasing application of ML in stock market prediction with better accuracy and adaptability.

B. *Types of Machine Learning Methods Deployed in Stock Market Prediction*

Supervised learning methods contain techniques useful in stock market prediction such as learning historical data and predicting future outcomes. Some of these methods include regression models, decision trees, support vector machines, and neural networks.

1) *Regression Models*

Linear Regression is one of the simplest and most widely used predictive methodologies in predicting stock prices. This simple linear regression models the dependence of differing variables, which is the price of a stock (dependent), on one or several independent variables known as features by fitting a linear equation that will best represent the data by minimizing a loss function (Murphy, 2012). Although much more sophisticated modelling could be done, even with this simple method, linear regression could work relatively well for short-term predictions, as it is often used as a benchmark to compare other more advanced models. Polynomial regression takes that a step further and fits the data with a polynomial equation, hence getting a flexible fit.

2) *Decision Trees and Random Forests*

Decision trees are a non-linear predictive modelling technique. The fundamental concept underlying the approach is, therefore, to split the data and make subsets depending on the value of input features and, hence, making a tree-like model of decisions. This easy-to-interpret algorithm can handle numerical and categorical data. As an ensemble method, a random forest builds many trees while training, since many trees develop the single decision process, reducing overfitting and enhancing accuracy overall by a collection of trees (Breiman, 2001).

3) *Support Vector Machines (SVM)*

SVMs are essentially powerful classifiers and regressors. In prediction domains of the stock market, SSVMs find the best line that separates different classes of data or fits to the data for regression. They are effective in very high dimensions and protect against overfitting extremely well, especially where the number of dimensions is greater than the number of samples (Cortes & Vapnik, 1995).

4) *Neural Networks*

a) Feedforward Neural Networks: Basic Structure and Applications

The simplest kind of NN is the Feedforward Neural Networks. It has an architecture of an input layer, a hidden layer, and an output layer. All vectors of neurons in each layer are pairwise fully connected to those in the subsequent layer without cycles. FNN are designed to minimize the error of prediction using backpropagation by adjusting weights associated to those connections (Zhang, 2003). Whereas, in the case of stock market prediction, the complex and nonlinear relationships between historical stock prices and other features can be similarly modelled using FNNs, which are quite effective for short-run forecasts but fail when handling sequential data and therefore deal with a limited time dependency.

b) Recurrent Neural Networks (RNNs) and Long Short-Term Memory Networks (LSTMs): Their Role in Handling Time Series Data

RNNs are designed for sequential data; formulation of a loop allows information to persist across the time steps. This feature makes RNNs ideal for working with time series predictions, typically temporally ordered data points, such as in stock market forecasting. Traditional RNNs exhibit several intrinsic problems to cope with the vanishing gradient problem, which dramatically hinders the learning of long-term dependencies. LSTMs exceed expectations further by using memory cells and gating capacities, both for the long-term dependencies and temporal patterns. For historic price movements and trading volumes, they are shown to outperform several traditional RNNs and other models across a variety of studies. As stated by Hochreiter and Schmidhuber (1997),

c) Ensemble Methods: Combining Multiple Models to Improve Prediction Accuracy

Ensemble methods are intended to enhance the effectiveness of prediction methods by combining many different machine learning models to take advantage of the strengths of each one of them and mitigate the weaknesses of those under consideration. Techniques like bagging, boosting, and stacking are widely used to predict stock market movement. An example of bagging is Random Forests, which is essentially a procedure that entails the training of a few models on different subsets of the data and taking the average of the predictions to reduce variance. Boosting, found in AdaBoost and Gradient Boosting Machines, trains models sequentially to correct the errors made by previously trained models, thereby improving precision. Stacking uses the predictions of the base models, where a meta-learner combines them to obtain the final prediction, which considers the output from all base models. All these ensemble approaches can capture many diverse patterns in the stock market data very effectively and hence are more robust and accurate in their forecast as compared to any single model.

5) Unsupervised Learning in Stock Market Prediction

a) Clustering Techniques: K-means, Hierarchical Clustering for Pattern Recognition in Stock Data

Clustering, unsupervised learning techniques, play a paramount role in identifying patterns and segmenting data into distinct groups with no predefined labels. K-means clustering, and hierarchical clustering are arguably the most popular techniques of clustering methods for stock market prediction.

b) K-means Clustering

K-means clustering breaks the data into different clusters based on their feature similarities, working by extending data points iteratively to the cluster of the nearest centroid and recalculating again the centroids as a mean point of positions of all points in every cluster. Repeat this process until it converges—in other words, the position of the centroids does not change significantly. K-means could be used with stock markets to segment stocks according to similar price change or volatility features. Using these cluster features, an analyst would be able to identify trends in certain sectors or form portfolios diversified on these characteristics.

c) Hierarchical Clustering

There are two ways in which hierarchical clustering creates a hierarchy of clusters: agglomerative and divisive. Agglomerative clustering takes the starting point of having each data point as its own cluster. The process proceeds iteratively, merging at every step the closest pairs of clusters until there is only one cluster left. On the other hand, divisive clustering starts with the entire dataset as the sole big cluster and recursively divides it into smaller clusters. The result of this process gives a dendrogram, which is a visual lineage of the data; it is easy to explain which data points keep which relation between them at different levels of granularity (Johnson, 1967). In the context of stock market prediction, hierarchical clustering helps to understand the nest structure of stock groups, identify the subsectors, and analyze the correlations within and between the clusters. Both K-means and hierarchical clustering are applied to discover hidden patterns in stock data, which helps in making more informed decisions in trading and investment, and strategic planning.

d) Dimensionality Reduction: PCA and Its Role in Feature Extraction and Noise Reduction

Dimensionality reduction techniques, like PCA, are very important in the simplification of complex data. They do this by reducing the number of features and at the same time preserving most of the variance from the original data. This will be quite useful in stock market prediction, for most datasets have several correlated features, hence redundancy and increased noise.

e) Principal Component Analysis (PCA)

One of the main properties of PCA is that it transforms the original features into a new set of uncorrelated variables called principal components. These components are ordered in a manner such that the first few contain most of the data's variance. Essentially, this is the calculation of eigenvectors and eigenvalues of the covariance matrix of the data. In 1986, Jolliffe explained that the eigenvectors represent the directions of maximum variance. Dimensionality is reduced because PCA projects the data onto the principal components, usually catching the essential structure of the data with less features.

In stock market prediction, PCA serves multiple purposes:

- Feature Extraction: PCA helps in the selection of only those principal components that contribute to maximum variance and hence picks the most informative features, reducing the dimensionality of the dataset without much loss of information. This simplifies the modelling process and enhances computational efficiency for the same.
- Noise Reduction: Since PCA contains only principal components that house key patterns in data, it filters out noise associated with less important components. This would result in cleaner

datasets and hence improve the performance as well as the accuracy of predictive models.

For example, in the case of many technical indicators with a dataset containing macro variables and sentiment scores, PCA will be able to identify the drivers of stock price movements, thereby reducing the dimensionality of the dataset without losing its predictive power. Thus, it will enhance the interpretability of the models as well as reduce potential risks of overfitting by the elimination of noisy or redundant features.

Thus, unsupervised learning techniques, particularly clustering and PCA, form an important component of stock market prediction. Clustering helps in arranging meaningful groups within data to recognize patterns and make some strategic analytical decisions. At the same time, PCA will reduce the dimensionality and extract only the essential features against noise, hence optimizing the performance of the predictive model.

6) Reinforcement Learning in Stock Market Prediction

a) Q-Learning and Deep Q-Learning: Strategies for Portfolio Management and Stock Trading

Reinforcement learning in the field of machine learning is that in which agents learn to make sequences of decisions through the interaction with an environment to maximize the cumulative rewards. One such very valuable application of this will be in the stock market prediction for portfolio management and in stock trading where the decisions should be changed continuously according to the market conditions.

b) Q-Learning

Q-Learning is one of the model-free RL algorithms whereby an agent learns some value function, $Q(s,a)$which approximately gives the expected utility for takin g action a in state s. These Q-values are updated iteratively by the agent based on the rewards received from the environment while trying to reach an optimal policy with the application of the Bellman equation. Q-Learning in portfolio management tries to resolve optimal actions like buy, sell, or hold, in respect to the present market state aiming at balancing a portfolio for maximizing return and minimizing risk.

c) Deep Q-Learning

Deep Q-Learning is an extension of Q-Learning using deep neural networks to approximate the Q-value function. It enables handling of large-dimensional state spaces typical in financial markets, (Mnih et al., 2015). The Deep Q-Network learns from raw market data itself and extracts complex patterns from it to make more informative trading decisions. By using techniques such as experience replay and target networks, it becomes stable during training and finally learns strategies that outperform the traditional methods themselves.

C. Evaluation of Machine Learning Methods

1) Model Performance Metrics

One can consider looking at performance when dealing with machine learning models, which would yield results of the reliability and accuracy in predicting the stock market. We will consider some of the main metrics:

a) Mean Absolute Error (MAE)

MAE gives a simple interpretation about how accurate the predictions are. It measures how far predicted values are from real values, given by their absolute average difference. The lesser the value that MAE returns, the better the model does.

b) Root Mean Square Error (RMSE)

RMSE is the same as the square root of the average squares of differences between the predicted and actual values. Being sensitive to outliers, it gives a notion of the magnitude of prediction errors. Low values indicate better accuracy.

c) Mean Squared Error (MSE)

The MSE is an average of the squares of the differences between predicted and real values. In comparison with MAE, it gives much more penalty for larger errors, so it's useful when one wants to understand the variance of prediction errors.

d) R-Squared (R^2)

R^2 is usually employed as an indicator of the amount of variance that the dependent variable can be accounted for based on the independent variables. The range occupies a space from 0 to 1, with the higher values indicating the good fitness of the model to the data.

2) Accuracy, precision, and recall of classification

These are the must-have metrics when the task is to predict categorical outcomes, such as up/down market directions.

Accuracy: measures the proportion of correct predictions out of total predictions.

Precision (Positive Predictive Value): is the ratio of the number of correctly predicted positive observations to that of total predicted, giving the percentage of correct predictions for positive instances and, therefore, the accuracy of the model.

Sensitivity (Recall): This is the proportion of the number of correctly predicted positive observations to that of all actual positive cases, and it gives a measure of how many actual positive cases the model can capture.

These are the metrics, among others, which give nuanced insight into model performance during fine-tuning and, subsequently, selecting the most appropriate ones for stock prediction.

D. Comparison Studies of Machine Learning Methods in Stock Market Prediction

1) Case Studies

Many of the important studies compare the performance of various machine learning methods in stock market prediction. For example, in the study by Patel et al. (2015), four machine learning algorithms—Artificial Neural Networks, Support Vector Machines, Random Forests, and Naive Bayes—, which are trained and tested with stock price data are compared. Their results indicated that Random Forest and SVM performed best among the compared models considering factors such as accuracy and robustness. Another important contribution came from a study by Fischer and Krauss, who, in 2018 conducted a piece of research comparing Long Short-Term Memory neural networks to traditional Machine Learning models; it showed that LSTMs were much better at modelling temporal dependencies and returned superior predictive performance.

a) Artificial Neural Networks

ANNs are very powerful in capturing any non-linear relationship but are prone to overfitting, especially when dealing with small datasets. They also require high computational resources for training.

b) Support Vector Machines

They work very well in high-dimensional spaces, robust to overfitting, and are computationally intensive. Moreover, their performance usually goes down with large datasets.

c) Random Forests

2) Random forests offer both high accuracy and robustness by considering multiple trees from decisions; they deal very well with Strengths and Weaknesses found big datasets. They can, however, turn out to be less interpretable as compared to the simpler models.

a) Naive Bayes

NB is computationally efficient and does very well for small datasets. The main limitation with this algorithm is the assumption of independence of features, many of which do not normally hold in stock market data.

b) Long Short-Term Memory (LSTM) Networks

LSTMs show very good results in time series prediction, being capable of learning even the longest dependencies. To be at their best, they require many computing resources and large datasets.

3) Contextual Performance

These methods will have their performance varied across different market conditions and data scenarios. During periods of low volatility, simple models such as SVM and NB can work decently; then, RF and LSTMs could grasp more complex patterns and temporal dependencies in volatile markets. The effectiveness of models will further depend on the quality and granularity of data. The LSTM sequential learning capabilities can capture more in high-frequency trading data, whereas the RF or SVM may be good special model family for the prediction of daily stock prices.

In summary, while no model may reign supreme in all situations, the choice of machine learning methods should be informed by whatever is peculiar to market conditions, available data, and computational resources.

E. Conclusion

The results of this literature review identify a considerable amount of progress realized toward stock market prediction through various machine learning techniques. Though foundational, traditional statistical models often miss the capture of complex, nonlinear relationships intrinsic to the financial data. Machine learning methods—including supervised, unsupervised, and reinforcement learning in the forms of regression models, decision trees, neural networks, clustering, PCA, and others—provide robust alternatives by effectively dealing with these complexities. ensemble methods that combine several models are of interest in improving predictive accuracy. Although all the approaches have their strengths, each of them has some limitations, such as computational intensity of neural networks. This review requires the correct choice of a model in view of specific market conditions and data characteristics, as well as available resources. Although there is no method that can be said to outperform others universally, appropriately strategized application of machine learning techniques could improve the accuracy and reliability of predictions relating to the stock market.

II. METHODOLOGY

A. Data Collection and Description

1) Data Source:

The historical stock price data in this study was obtained from yahoo finance via the yfinance Python library. Yahoo Finance is one of the most famous financial databases that contains the daily historical information on waranties, stock prices, dividends, and trading volume for several securities (Chatterjee et al., 2021; Orsel and Yamada, 2022).

Feature No.	Feature Name	Feature Description
1	Open	The opening price of the stock for the trading day.
2	High	The highest price reached by the stock during the trading day.
3	Low	The lowest price reached by the stock during the trading day.
4	Close	The closing price of the stock at the end of the trading day.
5	Adj Close	The adjusted closing price of the stock, accounting for dividends and splits.
6	Volume	The total number of shares traded during the day.
7	MA_20	The 20-day moving average of the closing price, capturing short-term trends.
8	MA_50	The 50-day moving average of the closing price, capturing medium-term trends.
9	Volatility	The rolling standard deviation of the closing price over the last 20 days.
10	Lag_1	The closing price from the previous trading day, used to capture momentum.
11	Lag_5	The closing price from five trading days ago, used to capture short-term momentum.
12	Lag_10	The closing price from ten trading days ago, used to capture medium-term momentum.

2) Date Range:

The timeframe ranges from January 1, 2020, to June 1, 2024, thus encompassing the period that was particularly active in terms of changes on the market and the post-pandemic recovery. This time frame makes it possible to train and test the models on different market conditions thus making them more reliable (Soni et al., 2022).

3) Features:

The dataset consists of the defining parameters that underpin any changes in the price of stock like Open, High, Low, Close, Adjusted Close, and Volume. Furthermore, the features MA_20, MA_50, Volatility and Lag_1, Lag_5, Lag_10 were integrated in to the model to specify trends, momentum and risk characteristics respectively (Bansal et al., 2022; Oyewole et al., 2024).

4) Data Preprocessing:

Data preprocessing involved replacing missing values, feature scaling done by using MinMaxScaler on the data, new feature creation based on market trends and price fluctuations. The further split the dataset into the training and testing set for the proper validation of the models as well (Obthong et als., 2020; Chatterjee et al., 2021).

B. Exploratory Data Analysis (EDA)

1) Visualizations:

During EDA, a variety of plots was generated to make important inferences about the data at hand. Closing prices were plotted as a line graph to observe overall trends and identify any abrupt and volatile price movements (Figure 1). In addition to this, a volume trend plot (Figure 2) was produced to identify general trends in trading activity that may have implications for price fluctuation. A correlation heatmap (Figure 3) was used to understand the correlation between features and these correlated features are often removed while building models as they cause multicollinearity. Further, a distribution plot of the AAPL closing prices (Figure 4) analyze the frequency distribution of the prices while box plot (Figure 5) helped in identifying outliers in the price.

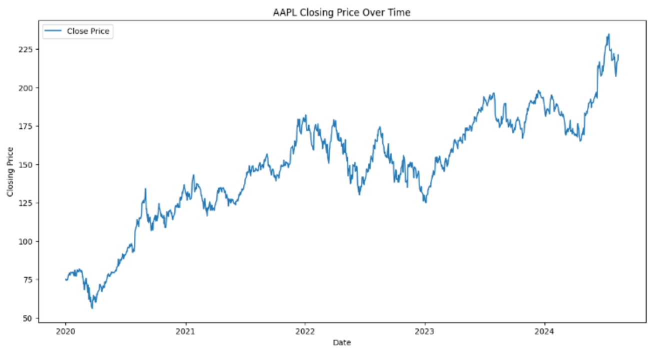

Figure 1: Line Plot of AAPL prices from January 2020 to June 2024

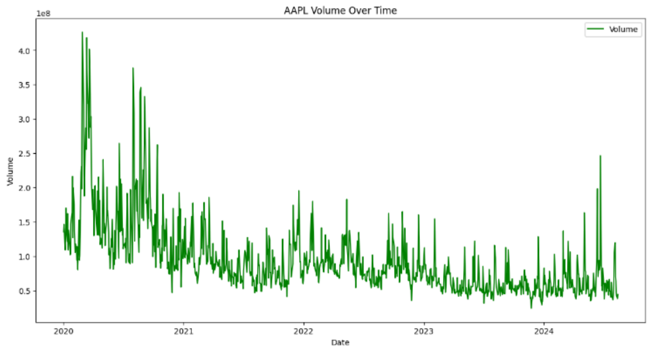

Figure 2: Volume of AAPL shares traded over time

Figure 2: Correlation heatmap of AAPL stock features, highlighting relationships between variables

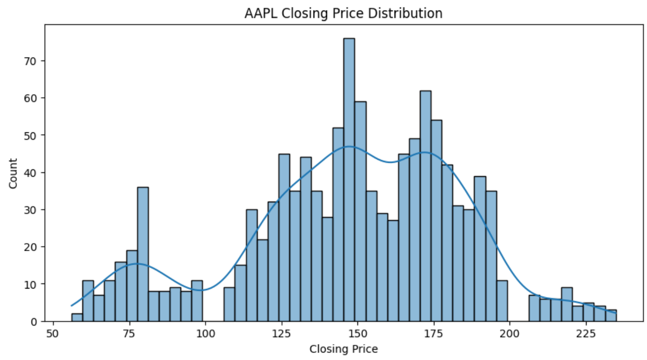

Figure 3: Distribution of AAPL closing prices

Figure 4: Boxplot of AAPL closing prices, indicating potential outliers

2) Insights:

The EDA provided some major findings as follows. The closing price line plot identified the time frames with high variability, which could be related to specific events in the economy during the research period (Mehtab et al., 2021; Darapaneni et al., 2022). The volume trend plot displayed those large volumes characterized the price movement anticipated the subsequent striking price changes, suggesting that volume could be considered a leading indicator of price movement (Chatterjee et al., 2021). The heatmap of the correlation matrix showed high levels of correlation between Open, High, Low, and Close prices that forced the elimination of some of these variables to avoid multicollinearity (Soni et al., 2022; Orsel and Yamada, 2022). The closing price distribution plot revealed that the prices were contained within certain intervals and depicted stability during some time frames as well as highlighted a few outliers, which could probably be a result of some key events or aberrations (Oyewole et al., 2024; Bansal et al., 2022).

C. *Feature Engineering*

1) Moving Averages:

Moving averages were thus developed to help in removing noises or short-term oscillations in the stock price information and to bring out the longer trends. Of them, the 20-day (MA_20) and 50-day (MA_50) moving averages were calculated. These forms of metrics give the short-term and the medium-term movements of the stock into consideration. Moving averages are very helpful when searching for the trends in stock price and the levels of support or resistance, which is helpful for the determination of future price (Mehtab et al., 2021; Darapaneni et al., 2022).

Volatility:

Fluctuations or volatility was computed by taking the 20-day moving standard deviation of the closing prices. This feature quantifies the fluctuations of stock prices, which is a critical measure of market risk. High volatility is usually interpreted as a state of flux or shifts in prices, especially in the opposite direction, which are excellent signs when trying to foresee stock prices (Chatterjee et al., 2021; Soni et al., 2022).

2) Lag Features:

The momentum factor is addressed by the lagged features, including Lag_1, Lag_5, and Lag_10. These features enable the models to analyze the historical price levels, which increases their accuracy (Orsel and Yamada, 2022). Based on the correlation matrix, features such as Open, High, Low, Adj Close, MA_20, Lag_1, and Lag_10 which are highly correlated were dropped off to eliminate multicollinearity.

3) Normalization/Scaling:

For normalization of the features, MinMaxScaler method was used which scales the features within the range of 0 and 1. This step is especially important for machine learning models, especially the neural network since it keeps all features balanced and it avoids situations where a given feature dominates the rest of the features because of its scale (Banerjee, 2020; Ratan, 2020).

D. *Model Selection and Justifications*

1) Models Chosen:

The set of models chosen for this study include Linear Regression, Decision Tree, Random Forest, Gradient Boosting Machine (GBM) and Long Short Term Memory (LSTM).

2) Rationale for Model Choice:

Consequently, Linear Regression was chosen as a baseline algorithm on purpose as this algorithm suits well the identification of linear correlations. Decision Tree was chosen for decision and interaction features because it is capable of prying apart non-linear relationships, unlike Decision Stump. Random Forest which is a composite of Decision Trees was used because it was accurate and had the ability to handle overfitting through the feature-based averaging. Amongst these models, Gradient Boosting Machine was selected as it can be capable of handling non-linear relationship in a way that learn from the erroneous model and by developing models progressively (Chatterjee et al., PG 2021; Soni et al., PG 2022). The LSTM networks for temporal data were used as the recurrent structures are very effective in capturing sequential dependencies that are very important for the stock prices time series forecasting (Mehtab et al., 2021; Darapaneni et al., 2022).

3) Handling Temporal Data:

Other options of models exist but the choice applied an LSTM model due to effectiveness it possesses when working with temporal data. LSTMs has the benefit of remembering sequences through the change in the gates used in building the model in contrast to most general machine learning algorithms. This makes it especially useful in this work since stock prices next price depend on the previous price (Oyewole et al., 2024). Lagged features and memorization characteristic allows LSTM to learn based on the time series; therefore, helps in the prediction of stock price data.

This encompasses these kinds of models: The statistical models give efficient explanation of the tendencies of stock prices since the models use econometrical methods to analyze the data of the stock prices and the machine learning models support the statistical models in such a way that the models discover as many patterns as possible in the data.

E. *Model Training and Evaluation*

1) Training Process:

In the training process, the data was divided into the training data and the test data where the training data constituted 80% of the total data and the test data constituted only 20% of the total data. This way, the models are also tested on data that has not been used for model training, thus giving a better measure of model efficiency. These models are Linear Regression, Decision Tree, Random Forest, GBM, and LSTM, and each to learn the underlying pattern in the data from the training dataset (Chatterjee et al., 2021; Soni et al., 2022).

2) Evaluation Metrics:

The accuracy of the models was assessed by evaluating Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R-squared (R^2). MAE and RMSE let identify how far from the real values prediction is on average, R^2 reveals to what extent the model can explain the variations of the target variable. All these metrics were selected since they give a detailed account of the model’s performance and the model’s ability to work well across different environments (Bansal et al., 2022; Oyewole et al., 2024).

F. *Model Interpretation Techniques*

1) LIME:

Interpretation of the discovered machine learning models was done using LIME (Local Interpretable Model-agnostic Explanations). LIME is based upon local interpretation of the model’s functioning where the function perturbs the input data and the changes in the predictions are examined. This technique is very useful when it comes to understanding more intricate models like the Random Forest as well as the GBM since they will assist us in pinpointing those features which impact on the model’s decision making when it comes to certain instance. There is significant importance to the interpretability of models and especially in the financial industry where the decision-making is made based on the models' predictions (Mehtab et al., 2021; Darapaneni et al., 2022).

2) SHAP:

SHAP (SHapley Additive exPlanations) measure was employed to measures the feature importance on the model's decision making. Interpretations based on SHAP values provide a global perspective to the features causing different type of influence on the entire set of data. As compared to LIME which only offers local explanations, SHAP offers a better picture on how the features and the interactions affect the result. This becomes more useful in models such as LSTM whereby accrual understanding of how different sequential features is important. With SHAP, we were able to achieve high detailing in prediction while at the same time having the possibility of determining how the models arrived at their outputs which optimized the decision-making process (Chatterjee et al., 2021; Obthong et al., 2020).

III. RESULTS

A. Model Performance Evaluation

1) Linear Regression:

Linear Regression formed the basic confirmation in this comparison. The performance metrics were as follows: Specifically, the results indicate that average MAE is equal to 0.0238, the level of Root Mean Squared Error (RMSE) was 0.0321, and coefficient of determination, R-squared (R^2) of 0.971. From these results, although Linear Regression successfully identifies the linear patterns in the data, it may fail to learn the other complex nonlinear patterns that are inherent in stock prices. From the high R^2, it can be deduced that the model fits in the context of explaining the variance of stock prices quite well, but the problem of linearity may lead to underfitting especially when complex patters are present (Chatterjee et al., 2021; Soni et al., 2022).

2) Decision Tree:

Decision Tree model had the lowest MAE of 0.0231, RMSE of 0.0342, and R^2 of 0.967. When it comes to the R^2 measure, the Decision Tree here is a little lower than Linear Regression, thus suggesting that it models the variance in the data slightly less effectively. Nevertheless, the Decision Trees algorithms work best when it comes to identifying non-linear interactions between the features. The RMSE is higher in Decision Tree and it might overfit as it considers noise data alongside the original data particularly in volatile markets such as stocks (Orsel and Yamada, 2022; Oyewole et al., 2024).

3) Random Forest:

The Random Forest model was also implemented and seemed to perform better with an MAE of 0.0175, RMSE of 0.0247 and R^2 of 0.983. Due to the lower variance in the Random Forest as it combines multiple Decision Trees and their results do not overfit like a single Decision Tree. Based on these findings, the ability of this model to perform even better and have the lowest RMSE of tree-based models reveal its better generalization to different market conditions (Chatterjee et al., 2021; Bansal et al., 2022).

4) Gradient Boosting Machine (GBM)

The performance of the GBM model was also quite good with an MAE of 0.0200, RMSE of 0.0272 and R^2 of 0.979. An important feature of GBM is that it gradually reduces errors – or, rather, it pays special attention to complex data that is difficult to predict. Logistic regression has a slightly low R^2 than Random Forest but the RMSE is almost similar meaning the model can accurately predict. Nevertheless, due to its complexity and computational demand, GBM is referred to outperform in terms of relative gain, especially when compared to a less complex and computationally expensive algorithm such as Random Forest (Mehtab et al., 2021; Obthong et al., 2020).

5) Long Short-Term Model (LSTM)

The LSTM model, developed to work with temporality established an MAE less than 0.0318, RMSE of 0.0406, and R^2 of 0.953. Even for time-series data, the LSTM poses relatively lower accuracy to the tree-based models of the same dataset. This achievement however shows how it is not easy to train deep learning models especially on comparatively small data sets that are most of the time associated with significant overfitting. The lower R^2 means that LSTM manages to model temporal dependencies, but at the same time, it needs more data or better fine-tuning of the model to match the performance of simpler models such as Random Forest and GBM (Mehtab et al., 2021; Darapaneni et al., 2022).

B. Results of Model Interpretation

1) LIME Interpretations

For every given model, a method known as LIME (Local Interpretable Model-agnostic Explanations) was used to understand the prediction. The LIME results evidenced the significance of features such as Lag_5 and MA_50 within the models throughout the implementation. For example, Lag_5 was the most influential feature in Random Forest and Gradient Boosting Machine, meaning that the recent dynamics of price is important for the future price. Likewise, MA_50, the 50-Day Moving Average was ensured to be an important feature, proving its role in capturing the medium-term price patterns of a stock (Mehtab et al, 2021; Chatterjee et al 2021). This aspect of LIME is instrumental in the financial domain specifically because it offers an explanation and, therefore, builds trust and acceptance of the model's predictions wherein the explanation can be linked to basic financial ratios.

Figure 5: LIME interpretation of Linear Regression model predictions

Figure 7: LIME interpretation of Random Forest model predictions

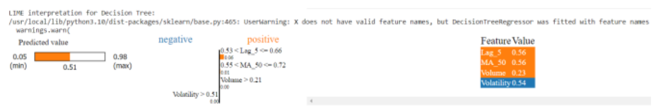

Figure 8: LIME interpretation of Decision Tree model predictions

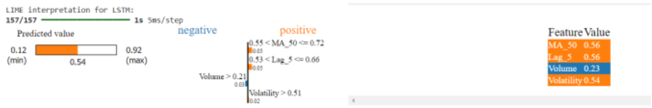

Figure 6: LIME interpretation of Gradient Boosting Machine model predictions

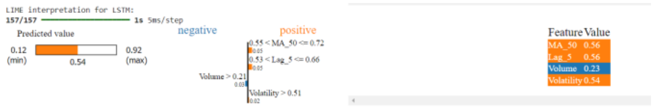

Figure 7: LIME interpretation of LSTM model predictions

2) SHAP Interpretations

This information complemented SHAP (SHapley Additive exPlanations) values to discuss feature importance further. These SHAP summary plots for all the models were relatively similar, and from all of them, Lag_5 and MA_50 featured as the most impactful variables. These were the features which had the highest SHAP values meaning that they were significantly influencing the model outputs. Among these features, Lag_5 was more outstanding in the LSTM model, which demonstrated the model's capability of learning temporal structures (Darapaneni et al., 2022; Orsel and Yamada, 2022). SHAP values gave a broader view compared to LIME, corroborating the conclusion that both Lag_5 and MA_50 are significant not only locally but also globally for the dataset.

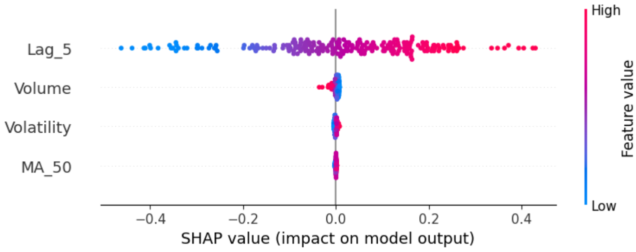

Figure 8: SHAP summary plot for Linear Regression model

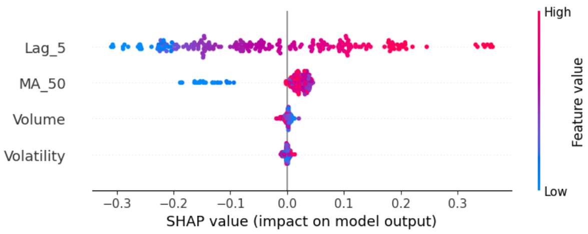

Figure 9: SHAP summary plot for Random Forest model

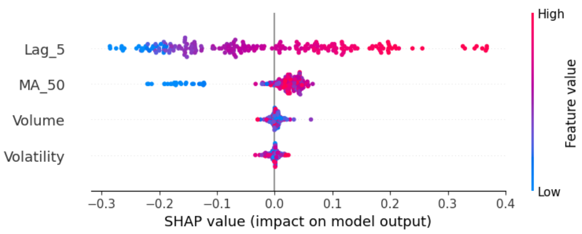

Figure 10: SHAP summary plot for Decision Tree model

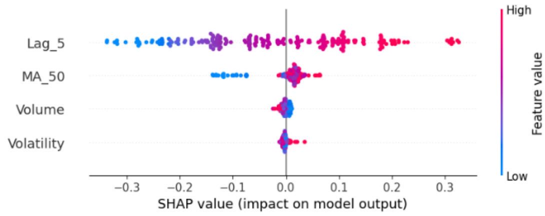

Figure 11: SHAP summary plot for GBM model

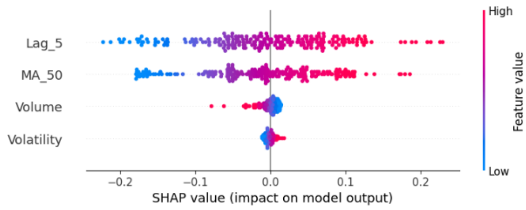

Figure 12: SHAP summary plot for LSTM model

C.

D. Discussion of Findings

1) Key Takeaways

The feature importance analysis showed that Lag_5 and MA_50 are the most critical features regardless of the model used, which means that the recent price fluctuations and medium-term moving average data should be pivotal when exploring stock prices. Besides, Random Forest and Gradient Boosting Machine (GBM) models showed lower MAE and RMSE values than the other models, which suggests these algorithms have a virtue to capture non-linear and complex relationships in data. The LSTM model which is otherwise effective for temporal data did not perform well in this case, implying that 'simple is better' principle may hold for stock price prediction in some cases, especially when the amount of available data is small (Mehtab et al., 2021; Chatterjee et al., 2021).

2) Implications:

From these analyses, Random Forest and GBM are good models for stock prices prediction, especially when used in ensemble with optimum features. That is why the fact that similar indicators and factors remain nearly identical across various models is indicative of the stability of these factors and, therefore, their effectiveness in predicting future price dynamics. This insight can be used to improve the essence of traditional and modern finance forecasting methods (Darapaneni et al., 2022; Orsel and Yamada, 2022).

3) Limitations:

The limitations of the study are that the data set used in this study is small and the LSTM model used is complicated indicating that perhaps larger data is needed for this LSTM model. Also, the models used historical data and it is still unclear what kind of performance they will provide in different markets (Soni et al., 2022; Oyewole et al., 2024).

E. Conclusion

1) Summary of Results

The study proved that with the ensemble models of Random Forest and GBM the results are even better as compared to other models and the most important features turned out to be Lag_5 and MA_50. Though LSTM was expected to perform well in case of temporal data, it was surpassed by simpler models which indicated that feature selection and model simplicity could be of higher importance in certain cases (Mehtab et al., 2021; Chatterjee et al., 2021).

2) *Future Work*

Future work may consider using more extensive data to improve the performance of deep learning algorithms such as LSTM. Moreover, it would be possible to enhance the quality of a model by including external data, for instance, news sentiment or macroeconomic factors. Finally, it may be viable to have more accurate predictions if techniques such as the transfer learning or the combination of the two or more algorithms is carried out (Bansal et al., 2022; Obthong et al., 2020).

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